Editorial Commentary: The Anterolateral Ligament Is Only Part of the Puzzle of Anterolateral Corner Injury in Anterior Cruciate Ligament Disruption and Reconstruction

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Abstract: Over the past decade, there has been an increased awareness of the role of the anterolateral corner (ALC) in the setting of anterior cruciate ligament (ACL) injury and ACL reconstruction. Although several studies have debated the existence, exact anatomy, and function of the ALC, the role in clinical practice is well recognized. Biomechanical studies have assessed the role of the anterolateral ligament (ALL), superficial and deep Kaplan fiber layers of the iliotibial band, the anterolateral capsule with the ALL, and the capsulo-osseous layer within the ALC. Furthermore, both biomechanical and clinical studies support the role of either ALL reconstruction or lateral extra-articular tenodesis, such as the modified Lemaire or Arnold-Coker modification of the McIntosh technique in the setting of ACL reconstruction. Not only have many randomized controlled trials shown decreased failure or rerupture rates with adding these procedures, but they also show minimal complication rates, equivalent to superior patient-reported outcome measures and decreased presence of persisting rotatory instability or positive pivot shift. These developments have caused a major shift in the treatment of young patients with ACL injuries, especially those who have ligamentous laxity, hyperextension, involvement in pivoting sports, or contralateral injuries. The next steps should focus on which patients require ALC procedures, the role of ALC procedures with hamstring versus quadriceps or bone—patellar tendon—bone autograft, and the development of post-traumatic osteoarthritis.

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In recent years, renewed anatomic interest[1] in the anterolateral corner (ALC) of the knee has led to extensive research in its components, particularly the anterolateral ligament (ALL). There has been considerable discussion about the existence, precise anatomy, and function of the ALL, particularly in the context of residual anterolateral rotatory instability (ALRI) after anterior cruciate ligament (ACL) injury and ACL rerupture rates.[2,3]

In the recent study titled "The Anterolateral Ligament Complex Has Limited Impact on Anterior Tibial Translation or Internal Rotation Stability in Anterior Cruciate Ligament-Deficient and Anterior Cruciate Ligament-Reconstructed Knees: A Systematic Review of Biomechanical Cadaver Studies," by Hohmann, Keough, Molepo, Arciero, and Imhoff,[4] the biomechanical role of the ALL is further investigated. The authors performed a thorough search, appropriately assessed the quality of the included studies, and applied multiple statistical tests to evaluate heterogeneity between the studies. A pooled meta-analysis of the data was not deemed feasible. Ultimately, 22 studies involving 284 specimens were included. Their systematic review concluded that there is no conclusive evidence that the ALL significantly limits anterior tibial translation or internal rotation stability in either ACL-intact or ACL-reconstructed knees under time-zero cadaveric biomechanical testing.

Although this systematic review provides additional insight into the role of the ALL, it addresses only part of the story. The ALC is a complex anatomic region, and its contribution to rotatory stability cannot be attributed solely to the ALL. Numerous studies have described the ALC as comprising the superficial and deep (Kaplan fiber) layers of the iliotibial band, the anterolateral capsule (including the ALL), and the capsulo-osseous layer.[5,6] All of these structures may be injured in conjunction

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with the ACL, as shown in various imaging studies, although the reported incidence of injury to individual components varied significantly.[7-10]

In our view, ALC injury encompasses a broad range of full or partial injuries to different anatomic structures, each contributing variably to ALRI after ACL injury. An ALC injury may consist of a partial tear of the anterolateral capsule, with or without involvement of the ALL, proximal and/or distal Kaplan fiber disruption, and/or injury to the capsulo-osseous layer. The cumulative effect of these injuries—alongside other concomitant injuries and individual knee chararacteristics[11]—determines the overall degree of ALRI in the ACL-injured knee.

It is, therefore, an oversimplification to focus solely on the ALL in the setting of ACL injury. The statements in the systematic review by Hohmann et al.[4]—that isolated ALL reconstruction or lateral extra-articular tenodesis may not be necessary in the management of ACL-deficient knees or may not provide biomechanical benefit when added to ACL reconstruction (ACLR)—should be interpreted with caution. A recent systematic review of biomechanical studies showed that all 6 included sectioning studies reported the iliotibial band to function as a secondary stabilizer to the ACL and to resist internal rotation. In contrast, only 2 of 6 sectioning studies found the ALL to contribute significantly to tibial internal rotation control.[12] Furthermore, this same study concluded that both a modified Lemaire tenodesis and an ALL reconstruction significantly reduce residual ALRI in isolated ACL-reconstructed knees, restoring internal rotation stability and internal rotation stability during pivot-shift testing.[9]

Clinically, the biomechanical relevance of combining ALC reconstruction with ACLR has been supported in high-risk, young patient populations. In the prospective study by the SANTI group, it was shown that adding ALL reconstruction to hamstring tendon graft ACLR resulted in a 3.1-fold reduction in ACL graft failure.[13] Furthermore, the STABILITY trial reported a statistically significant and clinically relevant reduction in graft rupture and persistent rotatory laxity 2 years postoperatively after adding a lateral extra-articular tenodesis to hamstring tendon ACLR.[14] More recently, several randomized controlled trials have shown similar dramatic reduction in failure rates, which further support treatment of the ALC in high-risk patients.[15,16]

In our clinical practices, we maintain a low threshold for adding ALC reconstruction to ACLR, particularly in high-risk patients. Since implementing this approach, we have observed a substantial reduction in ACL rerupture rates. The procedure is relatively straightforward and, when routinely performed, adds minimal time to the surgery without significant morbidity.[17]

Consequently, in our opinion, biomechanical analysis of injury to only 1 component of the ALC (i.e., the ALL) cannot and should not be directly translated to clinical practice.

Disclosures

All authors (W.A.v.d.W., J.P.v.d.L.) declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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